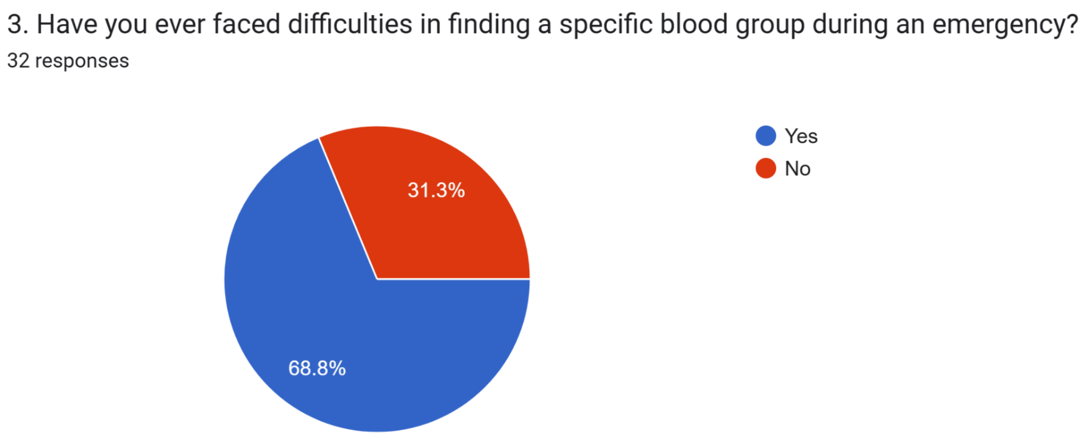
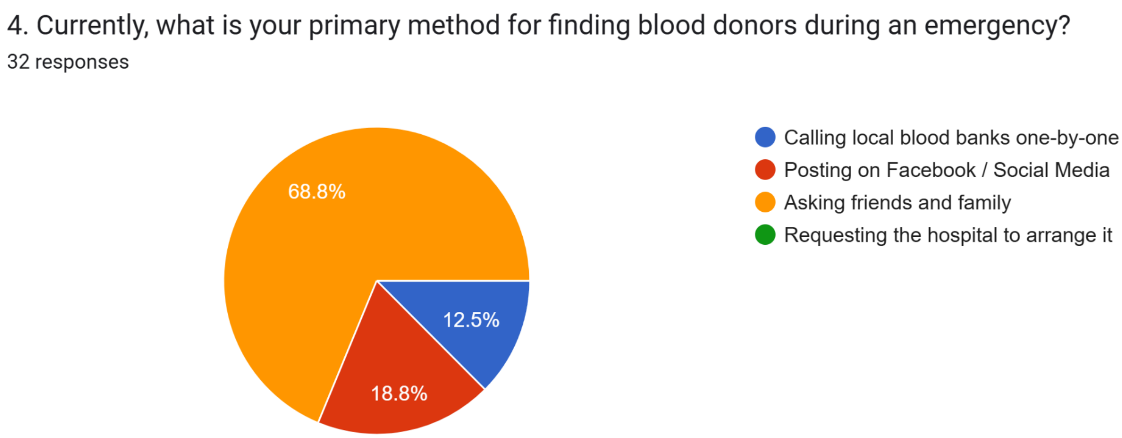
Problem Identification & Stakeholder Analysis

To determine the feasibility and necessity of a modernized Blood Bank Management System, a stakeholder survey was conducted with 32 participants (comprising donors and the general public). The results overwhelmingly indicated a failure in current emergency blood procurement. 68.8% of respondents stated they have faced difficulties finding specific blood groups during an emergency.

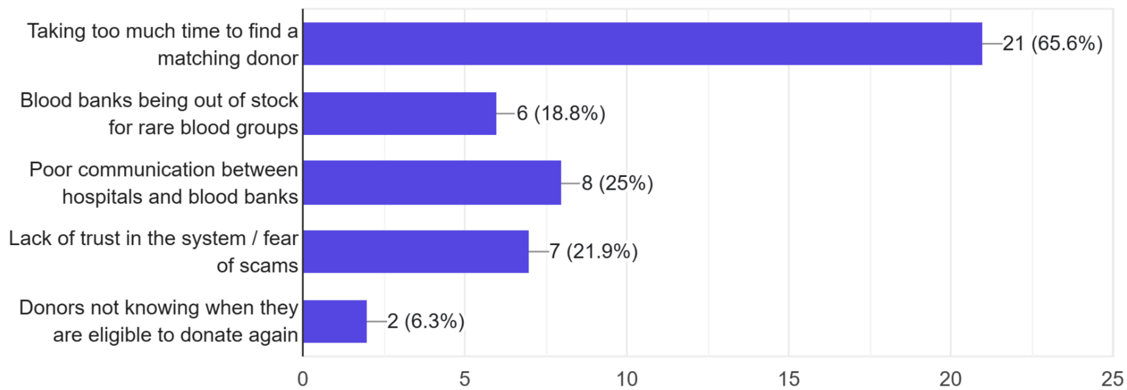


Furthermore, when trying to locate donors, 68.8% rely primarily on manually asking friends and family, while another 18.8% rely on Facebook and Social Media, rather than having access to an official medical database.



When asked about the biggest challenge in the current system, 65.6% identified "Taking too much time to find a matching donor."

5. In your experience, what is the biggest challenge in the current blood donation system?
32 responses

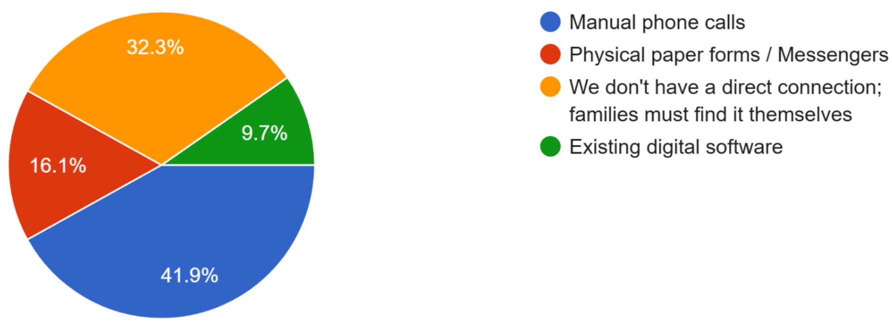


Relying on personal networks (friends and family) or social media algorithms during life-or-death emergencies is a critical system flaw that results in deadly delays. This explicitly justifies our architectural decision to build an AI Intelligent Matcher, which bypasses the slow process of asking around by instantly finding nearby eligible donors based on GPS and compatibility. When asked if they would prefer this AI system over relying on social media or their personal network, a staggering 96.9% said Yes.

2. Supply Chain & B2B Hospital Communication Failures

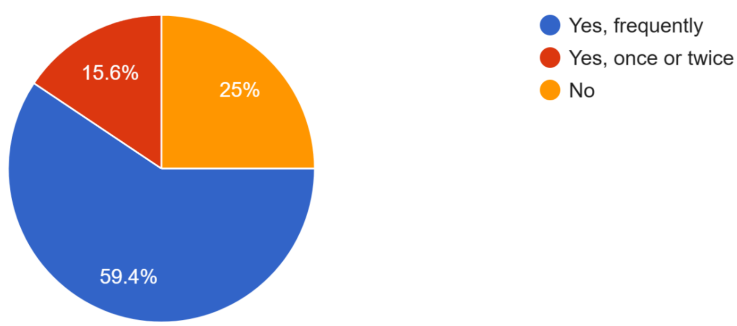
Our survey also exposed severe bottlenecks in how hospitals request blood from banks. 41.9% stated that manual phone calls are still the primary method of communication, and 32.3% noted there is no direct connection at all—leaving the burden entirely on the patient's family.

6. For Hospital Staff: How does your hospital currently request blood from external blood banks?
31 responses



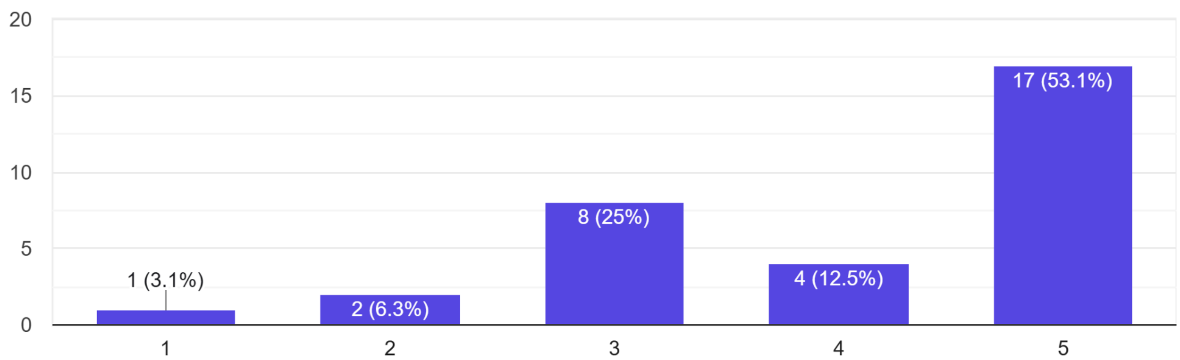
This lack of digital supply-chain integration leads to deadly stockouts. 59.4% of respondents reported frequently witnessing blood banks running out of specific groups during seasonal outbreaks like Dengue.

11. Have you ever witnessed a situation where a blood bank ran out of a specific blood group during a seasonal disease outbreak (e.g., Dengue)?
32 responses



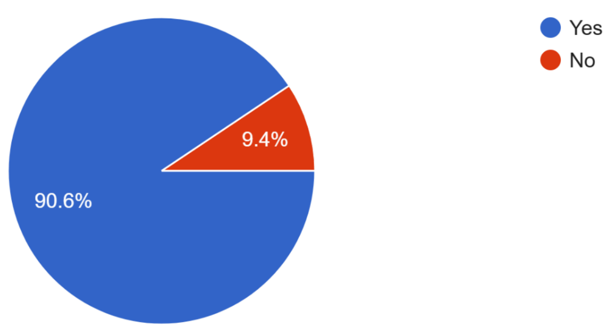
To resolve this, our system introduces a B2B Hospital Portal for seamless digital ordering, paired with an AI Demand Prediction Engine. When stakeholders were asked if software should be used to predict future shortages, over 53.1% rated predictive forecasting as Extremely Important.

12. How important do you think it is for blood banks to use software to predict future blood shortages before they happen?
32 responses



Furthermore, an overwhelming 90.6% of respondents agreed that human error plays a role when hospitals manually calculate how much blood inventory they need for the week, cementing the need for our automated AI prediction model.

13. Do you think human error plays a role when hospitals manually calculate how much blood inventory they need for the week?
32 responses

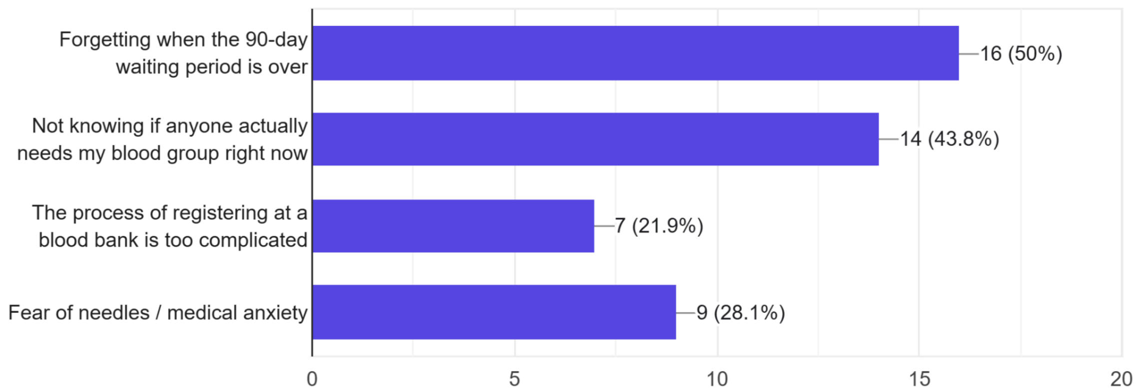


3. Donor Motivation and Security Concerns

Finally, we analyzed why existing donor retention is low. 50% of respondents stated they do not donate more often simply because they forget when their 90-day waiting period is over, and 43.8% said they don't know if anyone actually needs their blood.

14. What is the main reason you (or people you know) do not donate blood more often?

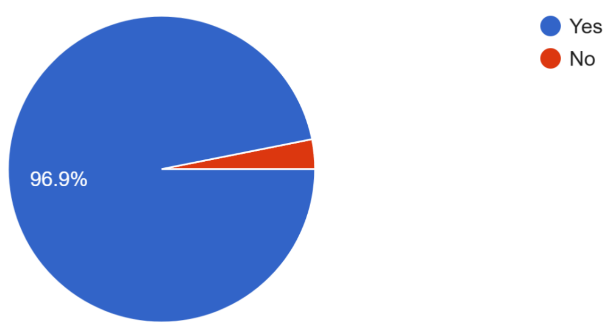
32 responses



If users received targeted, location-based automated alerts for emergencies, 96.9% stated they would be more likely to donate. This perfectly justifies our system's automated tracking and matching algorithms.

15. If you received an automated alert on your phone saying, "A hospital 5km away urgently needs your exact blood group," would you be more likely to donate?

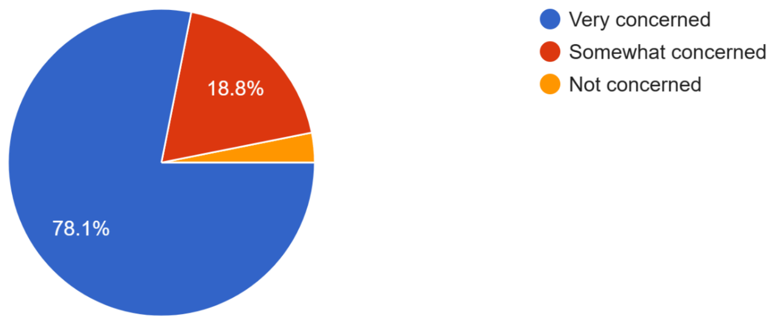
32 responses



However, security remains a barrier: 78.1% of users are very concerned about their medical and phone data being leaked by traditional paper registries.

16. How concerned are you about your personal data (phone number, medical history) being leaked by traditional paper-based blood bank registries?

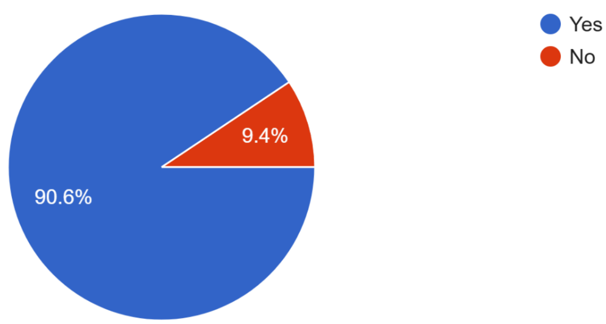
32 responses



This data strictly validates our implementation of strict Role-Based Access Control (RBAC) to hide private data, and the generation of secure Digital Donor ID Cards (with QR codes), which 90.6% of users agreed would make them feel more secure.

17. If the blood bank provided a secure "Digital Donor ID Card" with a scannable QR code, would this make you feel more secure about your medical data?

32 responses



Step 2: Write the Literature Review

2.1 The Inefficiencies of Traditional Blood Supply Chains

Traditional blood bank management heavily relies on manual record-keeping and telephone-based requisitions. Research evaluating blood supply chains highlights that manual systems suffer from high rates of human error, resulting in significant wastage of perishable blood units. In the foundational research on "Blood Bank Inventory Control" (Nahmias, S.), it is established that the lack of centralized, real-time databases leads to "information silos," causing severe delays during emergency situations. The transition to digital B2B hospital portals is critical to eliminate these manual communication failures.

2.2 The Role of AI in Demand Forecasting

One of the most complex challenges in blood bank operations is maintaining the delicate balance between supply and demand, as over-collection is just as detrimental as under-collection. The integration of Machine Learning (ML) and Artificial Intelligence (AI) allows systems to calculate future inventory requirements by analyzing historical data. According to recent research by Elmir, Hemmak, and Senouci (2023) in their paper "Smart Platform for Data Blood Bank Management: Forecasting Demand in Blood Supply Chain Using Machine Learning", the implementation of ML algorithms is essential. As they note, "forecasting demand in the blood supply chain using machine learning significantly improves the ability of blood banks to manage stock levels and avoid shortages." Our proposed architecture adopts this exact methodology through an AI Demand Prediction module.

2.3 B2B Integration and Role-Based Access Control (RBAC)

In modern system architecture, direct Business-to-Business (B2B) integration is essential for reducing friction in the supply chain. Integrating real-time usage feeds directly between hospital networks and blood banks must transition from reactive phone calls to proactive digital portals. However, exposing centralized medical databases requires stringent cybersecurity measures. The literature strongly supports the use of Role-Based Access Control (RBAC) to ensure that hospital personnel can only access requisition modules, while administrative staff retain full administrative privileges. This ensures both operational speed and data privacy.

2.4 Natural Language Processing (NLP) in Healthcare Interfaces

The adoption of complex management software is often hindered by steep learning curves for non-technical medical staff and donors. Recent advancements in Natural Language Processing (NLP) advocate for the implementation of conversational agents to solve this usability issue. As established by Laranjo et al. in their foundational paper, "Conversational agents in healthcare: a systematic review", conversational agents drastically improve user accessibility and can quickly execute complex queries using standard conversational language. By integrating an Action-Oriented Chat Assistant into our architecture, users can bypass nested user interfaces to instantly query blood stock or request donors, which is critical for saving time during medical emergencies.

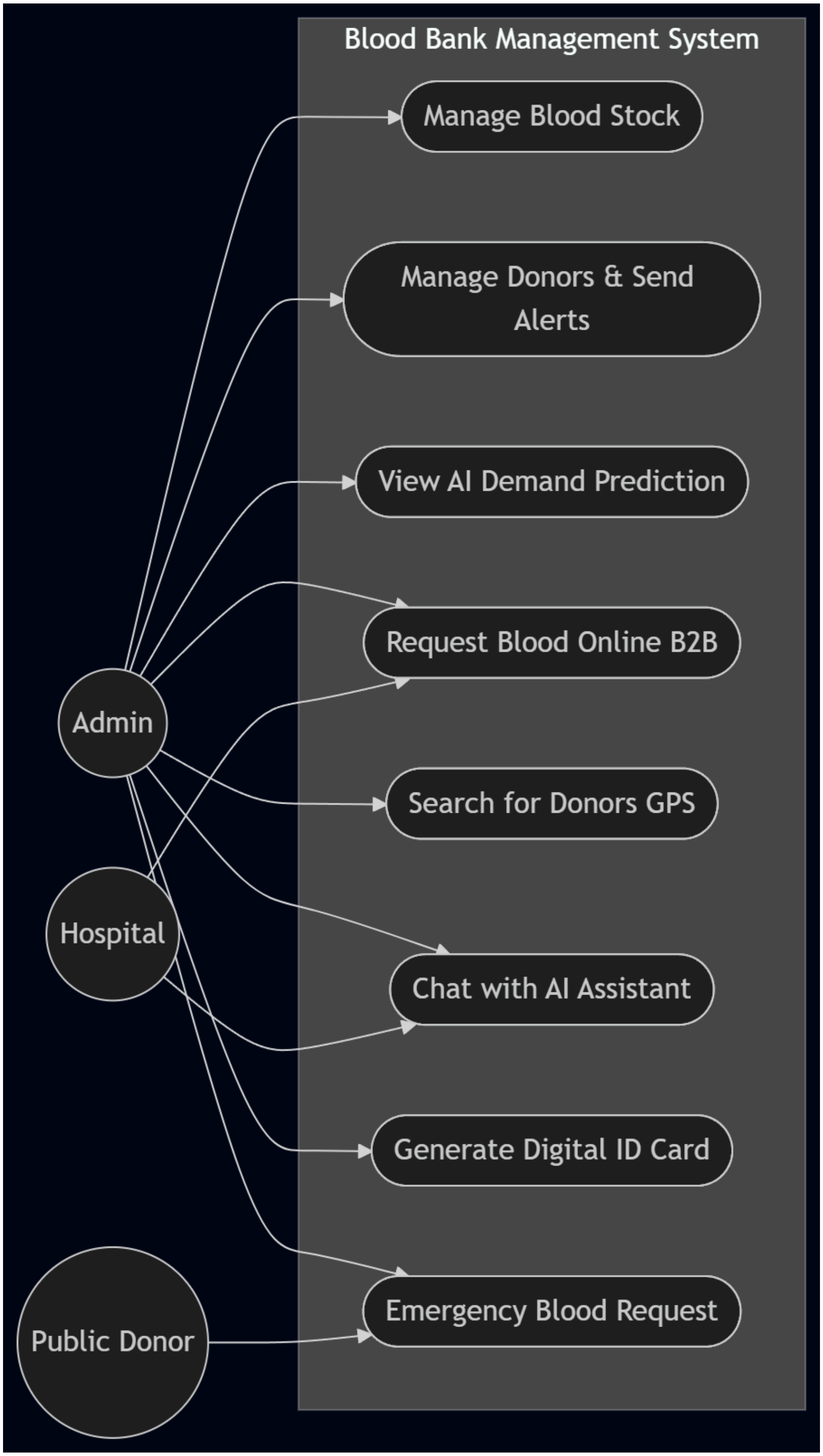
Bibliography (Real Citations for your report):

- Ben Elmir, W., Hemmak, A., & Senouci, B. (2023). "Smart Platform for Data Blood Bank Management: Forecasting Demand in Blood Supply Chain Using Machine Learning." (DOI: 10.3390/info14010031)
- Nahmias, S. "Blood Bank Inventory Control." (DOI: 10.1007/978-1-4419-7999-5_10)
- Laranjo, L., et al. (2018). "Conversational agents in healthcare: a systematic review." Journal of the American Medical Informatics Association. (DOI: 10.1093/jamia/ocy072)

Step 3: System Design, Models, and Cost Estimation

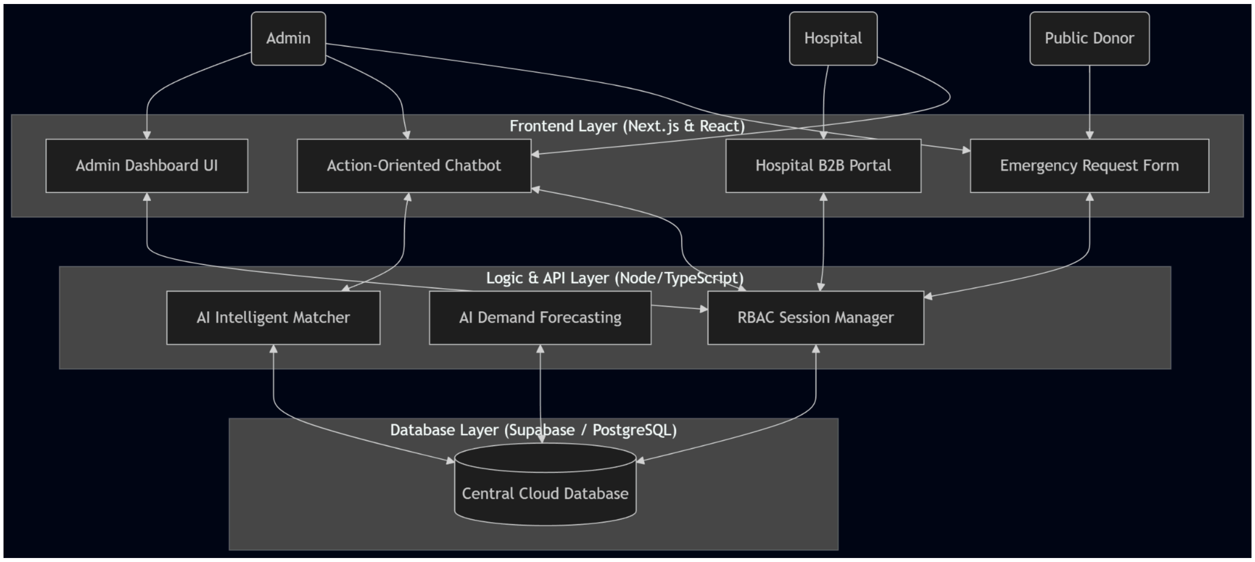
3.1 Use Case Diagram

This diagram outlines the three primary actors in your system and what they have permission to do (Role-Based Access Control). It now includes the Emergency feature and restricts Public Donors from the internal dashboard


Blood Bank Management System
Manage Blood Stock
Manage Donors & Send Alerts
View AI Demand Prediction
Request Blood Online B2B
Search for Donors GPS
Chat with AI Assistant
Generate Digital ID Card
Emergency Blood Request
Admin
Hospital
Public Donor

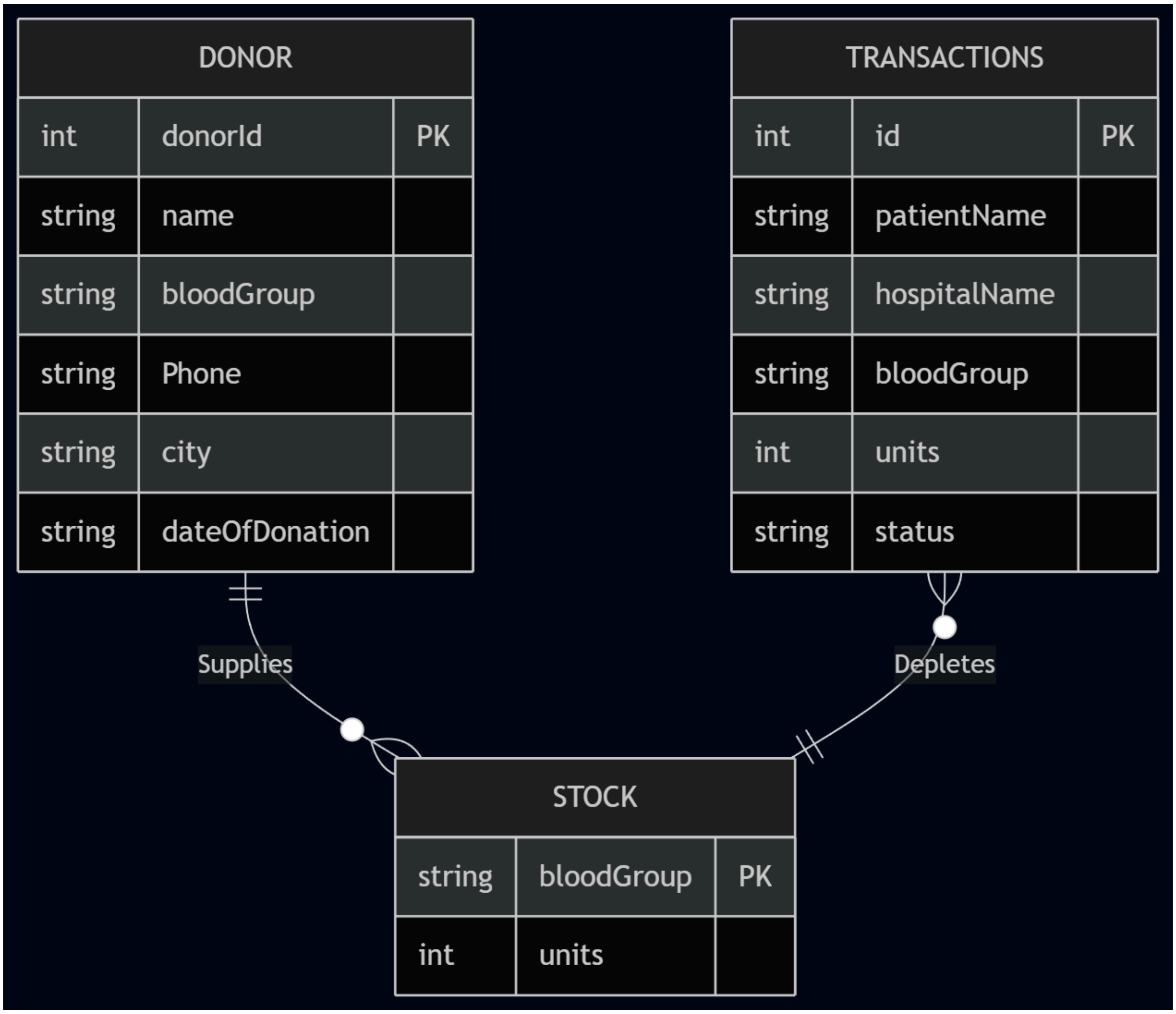
3.2 High-Level System Architecture

This diagram proves to your professor how the Chatbot, UI, and AI modules interact. (Note: Public Donors are completely restricted from the Dashboard and Chatbot to ensure data privacy. They can only access the Emergency Request Form).



3.3 Entity Relationship Diagram (ERD)

This represents the exact supabase_schema.sql database architecture we built for your project. There are three core tables that manage the entire system.



Time, Resource, and Cost Estimation

(You can copy and paste this text directly into the report)
System Estimations:

- Time to Develop: 4-6 weeks for an agile team (Phase 1: Database & RBAC, Phase 2: Frontend & Portals, Phase 3: AI Modules & Testing).
- Resources Required: 2 Full-Stack Developers (Next.js/React), 1 Cloud Database Architect (PostgreSQL/Supabase).
- Cost Estimation (Cloud Hosting): Deploying on Vercel (Frontend) and Supabase (Backend) allows the system to utilize a serverless architecture. The estimated running cost for up to 10,000 monthly active users is extremely low (under $50/month), making it highly feasible for low-budget municipal blood banks.

Step 4: Shortcomings & Technical Limitations

While the proposed Blood Bank Management System (BBMS) successfully resolves core inefficiencies in the blood supply chain, the current architectural design and prototype implementation possess several technical limitations that must be addressed before deployment in a real-world healthcare environment.

4.1 AI Module Constraints

Currently, the AI Demand Prediction Engine relies on heuristic algorithms and simulated historical data to forecast future blood requirements. A true production-ready machine learning model would require integration with a live data warehouse (e.g., Apache Spark or AWS Redshift) and continuous training on vast epidemiological datasets to accurately predict disease outbreaks (like Dengue). Furthermore, the Action-Oriented Chatbot utilizes a Regex and keyword-parsing engine rather than a true Large Language Model (LLM) backend (such as OpenAI or Llama-3). While the keyword parser is extremely fast and cost-effective, it struggles with highly complex or misspelled natural language queries.

4.2 GPS and Proximity Matching Limitations

The AI Intelligent Matcher currently pairs donors based on categorical data (e.g., matching a donor registered in "Dhaka" with a hospital in "Dhaka"). In a life-or-death emergency, relying on static city strings is a severe shortcoming. A production system must integrate live geolocation APIs (e.g., Google Maps API or Mapbox) to calculate exact routing distance, live traffic conditions, and the real-time geographic proximity of the donor's mobile device to the hospital.

4.3 Simulated Communications & SMS Gateways

To minimize development costs, the current architecture simulates automated donor notifications and emergency alerts through the User Interface. In reality, a blood bank system requires robust SMS and VoIP integrations (such as Twilio or Amazon SNS) to immediately push notifications to donors' cell phones when a match is found. Integrating these gateways introduces ongoing overhead costs that were excluded from the initial prototype scope.

4.4 Security and Public Access Restraints

Finally, to protect sensitive medical data and prevent the unauthorized scraping of donor phone numbers, the Role-Based Access Control (RBAC) completely restricts the general public from accessing the primary dashboard and chatbot. While this prevents data leaks, it limits the donor's ability to self-manage their profile or digitally interact with the system outside of the Emergency Request Form. A future iteration must decouple the architecture, creating an entirely isolated, highly secure Public Portal with its own authentication service (e.g., OAuth 2.0) separate from the internal Admin network.